

Photosynthesis

Photosynthesis is the process by which plants, algae, and some bacteria convert light energy into chemical energy stored in sugars. It is the foundation of nearly all life on Earth, supplying both food and the oxygen we breathe.

The overall reaction takes carbon dioxide and water and, using light energy, produces glucose and oxygen. Although this summary equation looks simple, it hides two distinct stages that take place in different parts of the chloroplast.

The first stage is the light-dependent reactions, which occur in the thylakoid membranes. Chlorophyll, the green pigment in the thylakoids, absorbs light, mainly in the blue and red parts of the spectrum, and reflects green light, which is why plants appear green. The absorbed energy excites electrons, which travel down an electron transport chain. This process splits water molecules, releasing oxygen as a by-product, and produces two energy-carrying molecules: ATP and NADPH.

The second stage is the light-independent reactions, also called the Calvin cycle, which take place in the stroma, the fluid surrounding the thylakoids. The Calvin cycle does not require light directly. Instead, it uses the ATP and NADPH made in the first stage to fix carbon dioxide into an organic molecule, ultimately building glucose. The enzyme rubisco catalyzes the crucial first step of carbon fixation.

Several factors limit the rate of photosynthesis. Light intensity, carbon dioxide concentration, and temperature each affect the process. When one of these factors is in short supply, it becomes the limiting factor that caps the overall rate, no matter how abundant the others are.

Photosynthesis and cellular respiration are complementary processes. Respiration releases the energy stored in glucose by consuming oxygen and producing carbon dioxide and water, effectively reversing the inputs and outputs of photosynthesis. Together they cycle carbon and energy through ecosystems.